

Polynomials, factor and remainder theorems - Set 1 - Solutions

Specification MM1.6.

Answer key

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Worked solutions

Question 1 POL-01 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

Answer: F

Remainder Theorem: $27 - 36 + 3k + 6 = -6 \Rightarrow 3k - 3 = -6 \Rightarrow k = -1$.

Full method - how and why

A remainder is given, so use the Remainder Theorem: substitute the root of the divisor and set the result equal to the remainder.

1 The divisor $(x - 3)$ is zero at $x = 3$.

why: The Remainder Theorem says the remainder on dividing by $(x - a)$ is $f(a)$; you need the value of x that makes the divisor vanish.

2 Substitute: $f(3) = 27 - 36 + 3k + 6 = 3k - 3$.

why: Work the numerical terms out first ($27 - 36 + 6 = -3$) so only the k term is left to carry.

3 Set equal to the given remainder: $3k - 3 = -6$.

why: This is what the Remainder Theorem gives you - an equation, not an identity.

4 Solve: $3k = -3$, so $k = -1$.

why: Simple linear equation.

Where it goes wrong

- Substituting $x = -3$ (the sign of the root of $x - 3$ is $+3$), which gives a different value of k .
- Setting $f(3) = 0$ out of habit - that is the Factor Theorem, and only applies when the remainder is zero.

Takeaway Divisor $(x - a) \Rightarrow$ substitute $x = a$; the result is the remainder, which is zero only when $(x - a)$ is a factor.

Question 2 POL-02 · Polynomials, factor and remainder theorems · spec MM1.6

$(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

Answer: F

Factor Theorem: $f(-2) = -16 + 4a + 10 - 6 = 4a - 12 = 0$, so $a = 3$.

Full method - how and why

'Is a factor' means remainder zero, so the Factor Theorem turns the statement into a single equation for a .

1 $(x + 2)$ vanishes at $x = -2$.

why: Write $x + 2 = x - (-2)$ if the sign is confusing; the root is the value that makes the bracket zero.

2 Substitute: $f(-2) = 2(-8) + a(4) - 5(-2) - 6 = -16 + 4a + 10 - 6$.

why: Careful with the odd power: $(-2)^3 = -8$, while $(-2)^2 = +4$.

3 Simplify: $4a - 12$.

why: $-16 + 10 - 6 = -12$.

4 Set to zero (factor $\Rightarrow$ remainder zero): $4a = 12$, so $a = 3$.

why: That is precisely the Factor Theorem.

Where it goes wrong

- Using $x = 2$ instead of $x = -2$, which gives $a = -3$ - an offered option.
- Sign slips on $(-2)^3$ and $-5(-2)$; write each term out separately.

Takeaway Factor $(x + k)$ means substitute $x = -k$ and set the result to zero.

Question 3 POL-03 · Polynomials, factor and remainder theorems · spec MM1.6

Fully factorise $x^3 - 7x + 6$.

Answer: B

$f(1) = 0$, so $(x - 1)$ is a factor: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6) = (x - 1)(x + 3)(x - 2)$.

Full method - how and why

Factorising a cubic: find one root by inspection, divide it out, then factorise the quadratic that remains.

- 1 Test small factors of the constant term 6: $f(1) = 1 - 7 + 6 = 0$, so $(x - 1)$ is a factor.

why: Any integer root must divide the constant term, so only $\pm 1, \pm 2, \pm 3, \pm 6$ are worth testing - and 1 is always the quickest to try.

- 2 Divide: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6)$.

why: Note the missing x^2 term in the original - treat it as $0x^2$ when dividing or comparing coefficients.

- 3 Factorise the quadratic: $x^2 + x - 6 = (x + 3)(x - 2)$.

why: Two numbers multiplying to -6 and adding to $+1$.

- 4 Full factorisation: $(x - 1)(x - 2)(x + 3)$.

why: Check by multiplying the constants: $(-1)(-2)(3) = 6$. ✓

Where it goes wrong

- Forgetting the missing x^2 term and producing the wrong quadratic factor.
- Sign errors giving $(x - 1)(x + 2)(x - 3)$, whose constant term is $+6$ too - always check the x coefficient as well.

Takeaway Root-hunt among the divisors of the constant term, divide out, then factorise what remains; check by multiplying the constants back.

Question 4 POL-04 · Polynomials, factor and remainder theorems · spec MM1.6

Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.

Answer: F

$$f\left(\frac{1}{2}\right) = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1 = \frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}.$$

Full method - how and why

A divisor of the form $ax - b$. The Remainder Theorem still applies, but the substitution value is $\frac{b}{a}$, not b .

- 1 $2x - 1 = 0$ at $x = \frac{1}{2}$.

why: The remainder on dividing by $(ax - b)$ is $f(b/a)$ - the root of the divisor, whatever the leading coefficient.

- 2 Substitute: $\left(\frac{1}{2}\right)^4 + 2\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right) + 1 = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1$.

why: Work each power out separately: $\frac{1}{16}, \frac{1}{8}$, then the linear and constant terms.

- 3 Put everything over 16: $\frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}$.

why: A common denominator prevents fraction slips; $\frac{2}{8} = \frac{4}{16}$ and $\frac{3}{2} = \frac{24}{16}$.

Where it goes wrong

- Substituting $x = 1$ or $x = 2$ instead of $\frac{1}{2}$.
- Arithmetic sign slips giving $+\frac{3}{16}$, which is offered.

Takeaway For divisor $ax - b$, evaluate at $x = \frac{b}{a}$; convert everything to a common denominator before adding.

Question 5 POL-05 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.

Answer: B

Write $p(x) = (x - 1)(x + 2)(x - r)$. The constant term is $(-1)(2)(-r) = 2r = 4$, so $r = 2$; the third factor is $(x - 2)$.

Full method - how and why

A cubic with two known factors has a third. The constant term alone pins it down - no division needed.

- 1 Write $p(x) = (x - 1)(x + 2)(x - r)$, where $(x - r)$ is the unknown factor.

why: The leading coefficient is 1, so the product of the three monic linear factors is the whole cubic.

- 2 Compare constant terms: the constant of the product is $(-1)(2)(-r) = 2r$, and $p(x)$ has constant term 4.

why: The constant term of a product of brackets is the product of their constant terms - much faster than expanding everything.

- 3 Solve: $2r = 4$, so $r = 2$ and the third factor is $(x - 2)$.

why: Check: $(-1)(2)(-2) = 4$ ✓.

Where it goes wrong

- Reading the third factor as $(x + 2)$ by forgetting the sign flip between the root r and the factor $(x - r)$.
- Trying to find a and b first - unnecessary work.

Takeaway Compare constant terms to find a missing factor: the product of the constants in the brackets equals the constant term.

Question 6 POL-07 · Polynomials, factor and remainder theorems · spec MM1.6

Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

Answer: C

$2x^3 - 3x^2 + 4x - 5 = (x - 1)(2x^2 - x + 3) - 2$; check the expansion

$2x^3 - x^2 + 3x - 2x^2 + x - 3 = 2x^3 - 3x^2 + 4x - 3$, then subtract 2 for the remainder.

Full method - how and why

A division question where only the quotient is wanted. Comparing coefficients is quicker and safer than long division.

- 1 Write $2x^3 - 3x^2 + 4x - 5 = (x - 1)(Ax^2 + Bx + C) + R$.

why: Dividing a cubic by a linear factor gives a quadratic quotient and a constant remainder.

- 2 Compare x^3 terms: $A = 2$.

why: Only one product contributes to the highest power, so the leading coefficient comes free.

- 3 Compare x^2 terms: $B - A = -3$, so $B = -1$.

why: The x^2 term comes from $x \cdot Bx$ and $-1 \cdot Ax^2$.

- 4 Compare x terms: $C - B = 4$, so $C = 3$. The quotient is $2x^2 - x + 3$.

why: The x term comes from $x \cdot C$ and $-1 \cdot Bx$. (The constant term then gives $R = -2$, but the question does not need it.)

Where it goes wrong

- Getting the sign of B wrong, giving $2x^2 + x + 3$ - which is offered.
- Reporting the remainder instead of the quotient.

Takeaway Comparing coefficients from the top down finds a quotient one coefficient at a time, with no long-division layout to get wrong.

Question 7 POL-11 · Polynomials, factor and remainder theorems · spec MM1.6

Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

Answer: A

$(x + 1)(x - 2) = x^2 - x - 2$; multiply by $(2x + 3)$: the x^2 term is $3x^2 - 2x^2 = x^2$, coefficient 1.

Full method - how and why

Only one coefficient is wanted, so do not expand everything: collect just the ways of making x^2 .

1 Multiply the first two brackets: $(x + 1)(x - 2) = x^2 - x - 2$.

why: Two brackets at a time keeps the bookkeeping small.

2 To get x^2 from $(x^2 - x - 2)(2x + 3)$, pair x^2 with 3, and $-x$ with $2x$.

why: An x^2 term arises from (degree 2)×(degree 0) and (degree 1)×(degree 1) - list the pairings rather than expanding.

3 Add the contributions: $3x^2 + (-x)(2x) = 3x^2 - 2x^2 = x^2$, so the coefficient is 1.

why: Only these two products can contribute.

Where it goes wrong

- Missing the $(-x)(2x)$ contribution and answering 3.
- Expanding everything under time pressure and losing a sign.

Takeaway For a single coefficient, list the pairs of terms whose degrees add to the one you want.

Question 8 POL-13 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following is a factor of $x^4 - 5x^2 + 4$?

Answer: E

$$x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2).$$

Full method - how and why

A quartic that is quadratic in x^2 . Factorise fully and read off which of the offered factors appears.

1 Substitute $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

why: Only even powers appear, so the substitution works.

2 Undo it: $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$.

why: Replace u by x^2 in each bracket.

3 Factorise each as a difference of two squares: $(x - 1)(x + 1)(x - 2)(x + 2)$.

why: $x^2 - a^2 = (x - a)(x + a)$ - do not stop at the quadratics.

4 Compare with the options: $(x + 2)$ appears.

why: Equivalently, a factor $(x - a)$ exists iff $f(a) = 0$: $f(-2) = 16 - 20 + 4 = 0$ ✓.

Where it goes wrong

- Choosing $x^2 + 1$ or $x^2 + 4$ - these are sums, not differences, and are not factors.
- Testing only positive values when checking $f(a) = 0$.

Takeaway Factorise as far as possible; a quick check of $f(a) = 0$ confirms any candidate factor $(x - a)$.

Question 9 POL-14 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.

Answer: E

$$p(x) = a(x-1)(x-2)(x-3); p(0) = -6a = 12 \Rightarrow a = -2. p(4) = -2 \cdot 3 \cdot 2 \cdot 1 = -12.$$

Full method - how and why

Three roots determine a cubic up to a constant multiple. One extra value fixes the multiple.

- 1 Write $p(x) = a(x-1)(x-2)(x-3)$.

why: Given roots, start from the factorised form; the leading coefficient a is the only freedom left.

- 2 Use $p(0) = 12$: $a(-1)(-2)(-3) = -6a = 12$, so $a = -2$.

why: Three negative factors give a negative product - a very common sign slip.

- 3 Evaluate $p(4) = -2(3)(2)(1) = -12$.

why: Substitute directly into the factorised form; no expansion needed.

Where it goes wrong

- Taking $a = 2$ from $6a = 12$ and answering $+12$.
- Assuming $a = 1$ because the question says 'a cubic' - the leading coefficient is not given.

Takeaway Roots give the shape, one extra point gives the scale; keep everything in factorised form.

Question 10 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$$p(1) = 1 + a + b + c, \text{ and } p(1) = (1-2)(1+1)(1-3) = (-1)(2)(-2) = 4. \text{ So } a + b + c = 3. \text{ (Directly: } (x-2)(x+1)(x-3) = x^3 - 4x^2 + x + 6.)$$

Full method - how and why

You could expand, but $a + b + c$ is the sum of the non-leading coefficients, which $p(1)$ delivers instantly.

- 1 Note $p(1) = 1 + a + b + c$, so $a + b + c = p(1) - 1$.

why: Substituting $x = 1$ turns every power into 1, so $p(1)$ is the sum of **all** coefficients including the leading 1.

- 2 Write the factorised form from the roots: $p(x) = (x-2)(x+1)(x-3)$.

why: Monic, because the given polynomial starts with x^3 .

- 3 Evaluate: $p(1) = (-1)(2)(-2) = 4$.

why: Two negatives and one positive give a positive.

- 4 So $a + b + c = 4 - 1 = 3$.

why: Remember to subtract the leading coefficient.

Where it goes wrong

- Answering 4 by forgetting the leading 1 in $p(1) = 1 + a + b + c$.
- Expanding and mis-collecting; the expansion is $x^3 - 4x^2 + x + 6$, so $-4 + 1 + 6 = 3$ ✓.

Takeaway $p(1)$ is the sum of all coefficients - a one-second route to questions like this.

Question 11 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x - 2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x - 2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Full method - how and why

The factor theorem gives one direction; a repeated factor is strictly more than a root.

- 1 If $(x - 2)^2$ is a factor, then $(x - 2)$ is a factor, so $p(2) = 0$: $Q \Rightarrow P$. **P is necessary.**

why: Factor theorem: $(x - a)$ is a factor exactly when $p(a) = 0$.

- 2 Is P sufficient? Take $p(x) = x - 2$: $p(2) = 0$, but $(x - 2)^2$ does not divide a degree-1 polynomial.

why: A single root is not a repeated root.

- 3 So P is necessary but not sufficient.

why: Note the direction: the weaker condition is necessary for the stronger one.

Where it goes wrong

- Answering 'sufficient but not necessary' - the common pattern, but reversed here.
- Thinking $p(2) = 0$ and $p'(2) = 0$ together would be needed; that is exactly the extra condition for a repeated root, and its absence is the point.

Takeaway $p(a) = 0 \Leftrightarrow (x - a)$ is a factor; a **repeated** factor also needs $p'(a) = 0$.

Question 12 POL-06 · Polynomials, factor and remainder theorems · spec MM1.6

When $p(x)$ is divided by $(x - 1)$ the remainder is 3, and when divided by $(x + 1)$ the remainder is -1 . Find the remainder when $p(x)$ is divided by $(x^2 - 1)$.

Answer: B

The remainder has degree < 2 : $ax + b$ with $a + b = 3$ and $-a + b = -1$. So $b = 1$, $a = 2$: remainder $2x + 1$.

Full method - how and why

Dividing by a **quadratic** leaves a remainder of degree at most 1. Write it as $ax + b$ and use the two given remainders to find a and b .

- 1 Write the division statement: $p(x) = (x^2 - 1)Q(x) + ax + b$.

why: The remainder always has lower degree than the divisor, so with a quadratic divisor it is linear (or constant).

- 2 Note $x^2 - 1 = (x - 1)(x + 1)$, so substituting $x = 1$ or $x = -1$ kills the $Q(x)$ term.

why: Choosing the roots of the divisor is what makes the unknown quotient disappear.

- 3 $x = 1$: $p(1) = a + b$. The Remainder Theorem gives $p(1) = 3$, so $a + b = 3$.

why: The remainder on dividing by $(x - 1)$ is $p(1)$, which is given as 3.

- 4 $x = -1$: $p(-1) = -a + b = -1$.

why: Same reasoning with the other root.

- 5 Solve: adding gives $2b = 2$, so $b = 1$ and $a = 2$. Remainder $2x + 1$.

why: Two linear equations in two unknowns.

Where it goes wrong

- Assuming the remainder is a constant and trying to force a single number.
- Sign error at $x = -1$: $a(-1) + b = -a + b$, not $a - b$, which would give $2x - 1$ (an offered option).

Takeaway Remainder on division by a degree- n divisor has degree $< n$; substitute the divisor's roots to find its coefficients.